Two Laps Home

What the NEXUS record found, what it is, and the one swap it keeps re-reading

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Abstract

The NEXUS record is a ledger of numbered entries, each written from live program output. This paper reads its first 165 entries end to end for two things: what the record found, and what it is. The new results fall into nine groups, from relation vocabularies on a twelve-vertex witness to tetrahedral chains and the size of a polyethylene coil. At every level the same object returns: a swap with no fixed point, and a reader that cannot see it. Beyond whatever continuous frame it must hold, exactly one bit is lost to the reader — the sheet — and there are two ways home — hold the bit as a frame, or run a loop whose lift is open, which reaches the partner after one lap and comes home after two.

C0 and C1 force the first instance, because NOT is the only fixed-point-free map of one bit. Each later instance is that bit re-addressed by the next reader. Four laws carry it: (A) a reader is the quotient by its kernel, and the kernel is its way home; (B) a free involution makes two sheets and an odd loop; (C) a reflection at one level is a half-turn at the next, so a lock opens with one more freedom; (D) at a fixed point of the swap, an even reader is blind to the odd channel at first order.

The recursion is literal. A counter is NOT called recursively through the carry, and each wheel is a double cover of the one below. The record's twisted ring is the binary cyclic group — the spin lift of the plain ring.

One prediction was sealed before it ran: the diamond mirror-pair **ladder**. Among the frame-held exchange windows of the tetrahedral lattice, a conformation and its mirror image are *not reached* from one another at freedom 1 — every branch traced from one closes without meeting the other, for all five seven-bond mirror pairs — and are *joined* at freedom 2, for all twenty-three native eight-bond mirror pairs, with the mirror acting freely along every join. One more freedom lifts the lock — Law C, in matter.

Read in frequency, the swap's two characters are the zero frequency and the Nyquist frequency of a two-element group, so "two laps home" is the Nyquist count of the swap: one sample aliases the sheet, two resolve it, and no number of samples through an invariant reader can. A sealed, self-contained notebook reruns 39 checks: 38 pass, and one sealed prediction missed and is logged.

Throughout, the routes are fixed. C0 and C1 are a standing paradox: the transformation graph cannot itself change, yet everything that exists is change *inside* it. Nothing enters from outside — a thing is already on a route, which is how it got here — and it compiles the moment it is read.

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Versions. Version 2 (§ 166) corrected the claim that a reader's whole kernel is one bit, reworded the freedom-1 result as computational non-reachability, replaced six notebook checks that did not test what they claimed, and added two results from review: the bridge between the twisted register and the native partner (Section 2) and the Nyquist reading of the swap (Section 5). Version 3 (§§ 167–168) adds the group that the register and the partner generate — at odd width, the signed symmetries of the n-gon (Section 2) — reproduces and sources the bond-angle value of Section 8, and rewords passages that were written for the working exchange rather than for an outside reader. Appendix C lists every correction.

1. The question and the short answer

The aim is to find everything new in the record, to pull far enough back to see what it actually is, and to look for recursion. The method is the one the record uses for every gap (*reverse the gap*): name the gap by what must be there, subtract what known mathematics already supplies, and read the remainder as a copy of an earlier structure.

Here the gap is one structure that every level of the record re-reads. What is subtracted is a great deal of standard mathematics — coboundaries with coefficients, covering spaces and their monodromy, the inclusion of O(d) in SO(d+1), the spin double cover, configuration spaces of graphs, and the matrices of Flory's rotational-isomeric-state theory. What remains is a single fact used over and over: the swap that Co/C1 force on one bit, read again at each level by the next reader.

The short answer. The record is a long study of readers that cannot see a swap. Every reader in it is a quotient, and every result is one of three things: the kernel, which is what the reader cannot see; a route back, which recovers it; or an obstruction, which shows that no route exists at the given freedom. The kernel always contains one bit, the sheet — the swap's two-element group; when the reader is continuous, the kernel also contains a continuous frame (a translation, a rigid motion), which is held rather than lost. The swap is always an involution with no fixed point. At each level it is re-addressed in one of two ways — it stays a symmetry acting freely (two sheets and a loop that needs two laps), or it becomes a motion one level up, a half-turn, whose own lift carries the swap again as a sign. Swap, half-turn, sign, swap: that is the recursion.

The fixed routes. None of this is a program being written. The ring is read-only: its addresses, its partner map, its difference reader are fixed for good. Co says a distinction exists; C1 says every distinction must continue. Together they are a paradox — the structure cannot change, because a route that changed would no longer be the route, yet existence *is* the continuation C1 demands. The resolution is that all change happens *inside* fixed routes. A thing does not enter the graph; it is already on a route, which is exactly how it came to be here, and it "compiles" — its next state is read off — the moment anything traverses it. The reader names the result afterward. This paper is an account of the shape those fixed routes have, and that shape is one swap, read at every level.

What is new here.

- The record-wide reading, with four laws and their instances in one place (Sections 3–5).
- The identification of the record's twisted ring with the binary cyclic group (Section 2).
- The bridge between the twisted register and the native partner: the rotation and the reflection of one dihedral quotient, with the partner as the reflection corrected by C1; and the group the two generate, which at odd width is the signed symmetries of the n-gon (Section 2).
- The Nyquist reading: two laps is the Nyquist count of the swap group (Section 5).
- The diamond mirror-pair **ladder**, sealed and run: not reached at freedom 1, joined at freedom 2, R free (Section 6).

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- The binding of § 141's open question by type (Section 7).
- The first matter number read as the record's own head-on-to-overhead map (Section 8).
- The inventory of new results across §§ 1–168, with statuses (Appendix A).

On SHA. Parts of the record (§§ 105–114, 123–130) and one of the syntheses [4] treat SHA-256. This paper keeps SHA where the record keeps it: a deterministic transformation inside the binary domain, whose digest is a *name* — a many-to-one reader of an equivalence class of internal states, not an input. Like any deterministic map, it can be run backward only when its full internal state is kept; recovering an input from a digest alone is not attempted. Nothing in this paper bears on any hash function's one-wayness.

2. The seed: one bit that must change

One bit. Co gives at least two states. C1 forbids a state that is its own continuation. Of the four maps of {0,1} to itself, exactly one has no fixed point: NOT (check 1.1). NOT is an involution: one application reaches the partner, two return. Everything below is this fact re-addressed.

A register. Feed the outgoing bit of an n-bit shift register back through NOT, and you get the one-bit Möbius loop τ (§ 140). After n steps every bit has passed the inverter once:

$\tau^n = C$, $\tau^{2n} = I$.

Every τ-orbit has even length dividing 2n and holds its own complement; the number of orbits is (1/2n) Σ ϕ(d)·2^(n/d), summed over the odd divisors d of n — Golomb's count, OEIS A000016 (check 1.2).

Two sheets of one reader. The difference reader D reads the cyclic adjacent XOR of a word. It is exactly two-to-one, with fibres {x, Cx} and kernel {0, 1…1} (check 1.3). It carries the plain ring σ onto the even-parity difference words, and read with one extra bit at the seam it carries the twisted ring τ onto the odd-parity words; in both cases the shift becomes the plain shift. So the twist is a parity, and the plain and twisted rings are the two sheets over one rotation of difference words; the reader never tells x from Cx.

The binary cyclic group. On the orbit of 0, the map $\tau^k(0) \mapsto e^{(i\pi k/n)}$ is an isomorphism $\langle\tau\rangle \cong \mathbb{Z}/2n$ with $C \mapsto -1$; the reader is two-to-one onto an n-cycle and reads the *square* of the phase (check 1.4). This is the extension $1 \to \mathbb{Z}/2 \to \mathbb{Z}/2n \to \mathbb{Z}/n \to 1$ — exactly what the spin double cover $SU(2) \to SO(3)$ gives for rotations about one axis. The mathematics is standard; the identification is the point: **the loop that C1 forces on a register is the spinor of its clock.** One lap of the plain clock is a sign on the twisted one, and two laps bring the sign home.

Two bits, three bits, and the counter. Written as matrices, the bit flip X and the sign Z are reflections; XZ is the quarter-turn and $(XZ)^2 = -I$ the half-turn (check 1.6). At n = 3 the swaps {I, P, C, PC} act on the orbit {001, 100, 110, 011} as the Klein group, and on its three nontrivial characters as the three half-turns of space; their SU(2) lifts i, j, k each square to −1 (check 1.7). Adding one to a k-bit counter is NOT called recursively through the carry (§ 84): dropping the top bit carries +1 mod 2^k onto +1 mod 2^{k-1}, two-to-one with kernel $\{0, 2^{k-1}\}$ (check 1.8). Each wheel of an odometer is a double cover of the wheel below; the carry is the way home between levels. Read downward, the words fixed by the half-turn (rotation by n/2) are exactly the doubled words uu — the layer below (check 1.9).

The register and the partner, bound. The twisted register τ and the record's native partner P (bit reversal J, or the complement on palindromes) are two faces of one dihedral structure, once the bridge is stated at the right level. Upstairs, on the 2^n words, $J\tau J = \tau^{-1}$ and C commutes with both, so τ and J generate a dihedral group of order 4n whose centre is {I, C} (check 7.2; the presentation is Gemini's). Down on the sheet quotient

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X/C it is the dihedral group of order 2n: τ becomes the rotation of the difference circle — GPT's twisted reader D_1 turns τ into the rotation of the difference word (check 7.1) — and J becomes the reflection. P is not in the upstairs group for n ≥ 3 (check 7.3). It is J corrected by C1: where J fixes a word, a palindrome, P takes the complement instead, because no state may be its own continuation. On the shape channel the correction is invisible — δ(Px) = J(δx) for every word (check 7.4) — and P and J induce the same map on X/C. So **the partner is the fixed-point-free lift of the shape's reversal, and τ is a lift of its rotation.**

The ring's address order is a third thing. The native successor is one 2^n-cycle while τ has order 2n, so the two are never conjugate beyond n = 2 (check 7.7; GPT). The principal τ-orbit nonetheless sits in the native ring in order: lap one climbs the representatives $2^k - 1$ at increasing addresses, lap two climbs their facing partners, and 1^n takes the last address (check 7.6; Gemini's observation, corrected at n = 2, 3, where the seam swap trades two states across the middle). One sealed prediction missed here: P is affine over $\mathbb{F}_2$ exactly when the palindromes form a hyperplane, which happens at n = 2 and 3 — not "never for n ≥ 3", as sealed (checks 7.5, 7.5b).

The group they generate (§ 167). Write H for the complement on the palindromes, so that P = HJ. The failure of the dihedral relation, $Q = (P\tau)^2 = H\tau^{-1}H\tau$, complements a word exactly where palindrome membership changes across the step $x \to \tau x$ (GPT). Its n rotations and the complement generate the kernel of the action of $G_n = \langle \tau, P \rangle$ on the sheet quotient, so $|G_n| = 2n \cdot 2^n$ for n = 3 and n ≥ 5, and 64 at n = 4, as Schreier–Sims confirms for n = 3..10. Every twisted orbit meets the palindromes in at most one sheet pair, and the extension of the dihedral group by the kernel splits exactly at odd n (Gemini). At odd width the group is the signed n-gon. On the principal orbit $\{\tau^k(o^n)\}$ the n sheet pairs are the vertices of an n-gon: τ rotates it and flips one sheet, at the wrap, and P reflects it and flips every sheet. Then $G_n \cong \mathbb{Z}/2 \wr D_n$, acting faithfully on that orbit. At even width the orbit ties the parity of the sheet flips to the parity of the rotation, so the image there has index 2, and for n ≥ 6 the rest of the group is a single central swap — the complement on the orbits that never meet a palindrome — which no homomorphism to $\mathbb{Z}/2$ detects (checks 167.3–167.10). The local flips are the partner's doing: the only elements of ⟨τ, J⟩ that fix every sheet pair are I and C (check 168.8), so the sheet can be flipped orbit by orbit only because C1 corrects J at the palindromes. The chords $(P\tau^s)^2$, products of s consecutive rotations of Q, flip two sheet pairs on each orbit that meets a palindrome — the palindrome and the point s steps before it — or none (§ 168).

3. What the record found

The new results of §§ 1–165 fall into nine groups; Appendix A lists them with statuses. Table 1 reads each level for the same five things: the swap, what the reader keeps, the bit it loses, the way home through a held frame, and the way home through an odd loop.

level (sections)	the swap	reader keeps	lost	home by a held frame	home by an odd loop
bit (§ 140)	NOT	nothing	the value	keep the value	NOT twice
ring (§§ 127–143)	$C = \tau^n$	the difference word	the face	one absolute bit	n more steps of τ
readings (§§ 136–137)	the antipode	RP^{n-1}	the sheet	a chosen hemisphere	an antipodal path
tokens (§§ 73–84)	label exchange	the unlabelled configuration	which token	hold one label	the exchange loop; 2[z]=0

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level (sections)	the swap	reader keeps	lost	home by a held frame	home by an odd loop
operators (§§ 144–147)	U_C	$H_+ \oplus H_-$	the mirror sector	a parity basis	holonomy −1 on an odd loop
planar chain (§§ 145–158)	the mirror C	the dihedral word	the face	the first bond, a rigid half-turn	the alternating route, frame turned by π
spatial chain (§§ 163–165)	the mirror R (improper)	the distance matrix	handedness	none (R is not a motion)	an internal loop, one more free bond
spin (R4)	−1 in SU(2)	SO(3)	the sign	none	a second full turn

Table 1. One structure, read at every level. In every row the swap acts freely and the discrete part of the reader's kernel is one bit, the sheet; where the reader is geometric (the two chain rows), a continuous frame of rigid motions is held as well.

The nine groups, in brief (Appendix A has the rows):

- **Relation vocabularies and readers (§§ 1–98).** A reader is a functional: every measured number is ⟨m(V), f_H⟩, a multiplicity vector paired with a reader that counts fixed vectors, so a zero has three causes — absence, a flat reader, a deficient vocabulary. The minus sector is the Nyquist band of a central involution; the two-token exchange class is the $\mathbb{Z}/2$ of non-planar graph braid groups. The swap is already here: "swap twice disappears" is $2[z] = 0$ in H_1.
- **Bits, fibres and routes (§§ 99–114).** The half-adder loses exactly w/2 bits and every fibre is one swap orbit; a name is a collision of two routes; a digest is a name, not unique data (DEAD END, routed).
- **Shells (§§ 99–118).** Capacity $2n^2$ arises with nothing supplied: $2\ell+1$ is the surface of the 2ℓ-bit layer, and the remaining 2 is the bit itself.
- **Rings, BBP and zero routes (§§ 116–134).** The register stack sums to 2^{64}; layer n/2 is the half-turn fixed set of layer n; eight BBP lanes carry seven constants and one zero route.
- **The partner ring and One Inverter (§§ 135–143).** The native partner has no fixed point; partial readings form a sphere with C its antipode; one feedback rule survives the filtration — the complement; one question is left open, which interaction supplies the single net inversion.
- **Operators and the honeycomb compiler (§§ 144–147).** C is the one energy-preserving operator; H splits into $H_+ \oplus H_-$; an odd quotient loop exists exactly when its lift holds x and Cx.
- **Gates, locks and leaving the plane (§§ 148–158).** The planar mirror is a proper half-turn of space; every planar lock opens in space (certified ≥ 0.977).
- **Space, contact and rigidity (§§ 160–162).** Head-on to overhead is the running product of increments with a held frame; every planar contact is invisible at first order.
- **Tetrahedral matter (§§ 163–165).** The diamond exchanges come with no search; 16 of 38 seven-bond classes connect; the ladder of Section 6; polyethylene's characteristic ratio from the same running product.

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4. Four laws

4.1 Law A — a reader is the quotient by its kernel, and the kernel is its way home

A reader that is a homomorphism identifies exactly the points differing by its kernel; it is the quotient by that kernel, and what it cannot recover is one kernel element, the held frame. Three instances: over $\mathbb{F}_2$ the kernel of the difference reader is {0, C} (check 2.1); over $\mathbb{R}$ the incidence map of a connected graph kills exactly the constants, and B^TB has one zero mode — the translation of a spring network (check 2.2); over SE(3) a rigid motion changes no internal coordinate, and the chain rebuilds from its torsions and its held frame (check 2.3). The kernel splits in two: a continuous part, the frame, which a reader must hold (a translation over $\mathbb{R}$, a rigid motion over SE(3)), and a discrete part, which across the record is the one swap, the sheet. Over $\mathbb{F}_2$ the kernel is all sheet; over $\mathbb{R}$ it is all frame; on chains in space it is both. Read down a chain the readers stack, each losing one more frame — positions to bonds loses a translation, bonds to turns a rotation, turns to the dihedral word the face. That is § 160's head-on-to-overhead mechanism, $G_n = G_0 g_1 \cdots g_n$, read as Law A three times: every integration needs one constant, and the constants are the frame.

4.2 Law B — a free involution makes two sheets and an odd loop

Let R be a fixed-point-free involution of a space M. Then $M \to M/\langle R \rangle$ is a double cover, and a loop downstairs lifts to a path from x to Rx exactly when its monodromy is odd — which happens in a component exactly when that component holds both x and Rx. After an odd loop the way home is a second lap. This is covering-space theory; what the record adds is the list of places it holds: the sphere of partial readings, with C the antipode (check 3.1); the honeycomb move Hamiltonian, which commutes with U_C and splits into two sectors; two tokens on a graph, where H_1 of the unordered configuration space has a $\mathbb{Z}/2$ summand exactly on the non-planar graphs (check 3.5; Ko–Park; Harrison–Keating–Robbins–Sawicki); the diamond window, where R fixes nothing and the two sheets join (Section 6); and SU(2), where one full torsion turn is −1 and two are +1 (check 3.9).

4.3 Law C — a reflection at one level is a half-turn at the next

A reflection of $\mathbb{R}^d$ in a hyperplane is the restriction of the rotation by π of $\mathbb{R}^{d+1}$ about that hyperplane. So a swap that is not a motion at level d becomes a motion at level d+1; a lock whose two sides are mirror images opens with one more freedom. The instances, up the tower: the bit flip's two reflections compose to the half-turn −I (check 1.6); the planar mirror is the proper half-turn diag(1, −1, −1) of space about the first bond (check 4.1), and all 46 planar locks open in space; two tokens give a braid in the plane and a sign in space, the graph form of the statistics of identical particles (check 3.5; Leinaas–Myrheim; Laidlaw–DeWitt); and the tetrahedral chain, whose improper mirror no rigid motion of the held chain supplies, so the way home must be internal — the ladder of Section 6.

4.4 Law D — at a fixed point of the swap, an even reader is blind to the odd channel at first order

If $f \circ R = f$ and $RX = X$, then at X the derivatives satisfy $df \circ dR = df$, so df vanishes on the −1 eigenspace of dR; the second-order form decides. A planar chain is fixed by the reflection through its plane, so every distance read from it is even under that reflection and its first variation along any out-of-plane torsion vanishes. Perturbing one torsion of a planar chain changes the squared distances only at second order — a tenfold smaller step shrinks the change exactly a hundredfold — while at a generic chain the change is first order, tenfold (check 5.1, with that control). Every planar contact is invisible at first order (§ 161). The diamond's singular endpoints all sit on a flat *tt* stretch, locally planar, where the same blindness returns (check 5.3). At

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the start of the record the same law appeared as sampling: a reader that averages over a central involution reads the zero-frequency bin and sends the minus sector to zero (§§ 35–37).

5. The recursion

Pulled back, the record has two towers, both built from the one swap.

Up: a tower of double covers. Each level is a double cover of the one below, the swap its deck transformation, the new coordinate the sheet bit: the counter $\mathbb{Z}/2^k \to \mathbb{Z}/2^{k-1}$ with the carry as the way home (check 1.8); the twisted ring over the plain ring (checks 1.3–1.4); SU(2) over SO(3) (check 3.9); the labelled token arena over the unlabelled one (§ 81); the diamond surface over its quotient by R (Section 6). Two laps below make one lap above.

Down: a tower of fixed sets. Each level is the fixed set of the swap one level up: layer n/2 inside layer n, under the half-turn (check 1.9); the plane inside space, under the mirror (Law D). At a fixed set an even reader is blind to the odd channel, and the second order decides.

Inward: rings inside rings. The centre word nests, $C_{n+2} = 1 \cdot C_n \cdot 1$ (§ 135); the odd-lane lift runs a smaller ring inside a larger one with its methods intact (§§ 141, 143). A smaller compiler keeps its swap inside a larger one — the fixed routes contain fixed routes.

Across, in frequency: two laps is Nyquist. The author's reading is that this is why Nyquist exists. Made exact, it is this. A free involution R generates a two-element group, and the two characters of a two-element group are its zero frequency and its Nyquist frequency, the alternating sign. Any observable f on an orbit {x, Rx} is a two-point signal, with a DC part (f(x) + f(Rx))/2 and a Nyquist part (f(x) − f(Rx))/2. One sample — one sheet, one lap — cannot tell them apart: that is aliasing. A reader invariant under R keeps only the DC part. Two samples — both sheets, two laps — are necessary and sufficient: the Nyquist count for the group $\mathbb{Z}/2$ is its order, two.

On the twisted ring $R = C = \tau^n$ is the half-period shift of the 2n-step clock, and the half-period shift sorts frequencies by parity: the odd-harmonic part of any signal is exactly its Nyquist part, and the even-harmonic part its DC part (check 7.10). So every state channel lives on the odd harmonics and GPT's twisted difference reader on the even ones (check 7.8). The reader is quadratic — it multiplies neighbouring signs of the anti-periodic extension — and odd plus odd is even: "the reader reads the square of the phase" (Section 2) is this statement in frequency (check 7.9). More samples through the invariant reader never restore the sheet. The whole two-lap window of D_1 is the same for x and Cx, while one face bit separates them (check 7.11, correcting Gemini). The sheet bit at every level of Table 1 is the Nyquist band of that level's swap, and § 37's finding that the minus sector is the Nyquist band of the central involution is the same statement at the start of the record.

What this does not claim: the factor 2 in the classical sampling theorem has its own derivation, from the two-sided band of a real signal. That it, too, comes from a two-element symmetry — frequency negation — is a reading of type, not a derivation.

The method recurses too. *Reverse-the-gap's* own grammar says a write carries its own inverse and that write and read are offset, never concurrent. That is the twisted ring: the bit written at step t is read at step t+n as its complement and comes home at t+2n, the register width being the offset. The record found the shape of its method in its first machine.

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6. The sealed run of § 165: the mirror-pair ladder

The seal (`73b2c9ef`..., Appendix B) was hashed at 2026-10-24T12:46:46Z, before the run. It re-seals a prediction whose original seals (`65eec639`..., `bfc5ed90`...) and outputs were lost when the computing environment was reset; the diamond enumeration is deterministic and reproduced the earlier orbit counts and mirror-pair indices exactly, which is why the re-seal could be made with confidence (Appendix C).

The windows. A § 163 exchange window holds four frame beads fixed (0, 1, k+1, k+2), has k arc bonds and freedom k − 4. Call (A, B) a **mirror pair** when a coordinate transposition p ∈ {x↔y, y↔z, x↔z} maps A to B exactly, fixing the four frame beads; then R = p is a free involution of the window's configuration space exactly when no configuration is p-symmetric. The enumeration is exact: at k = 5 the mirror pairs are orbits 14, 16, 29, 31, 37 of the 38 exchange orbits; at k = 6 they are 23 of the 175 (check 4.3).

The ladder. The prediction, sealed before the run:

- **Freedom 1 (k = 5): not reached.** For every one of the five seven-bond mirror pairs, the real-branch tracer of § 163 — started at A, following every branch through lattice crossings until it closes — never reaches the mirror image B, and every traced branch closed (check 4.4a). All 5 landed as sealed. This is computational non-reachability within the tracer, not a proof that the components are disjoint: GPT's caution stands, and a proof needs an exact decomposition of the real curve.
- **Freedom 2 (k = 6): joined.** Every one of the 23 native eight-bond mirror pairs is joined — a continuous frame-holding motion connects A to B = R(A), with the window Jacobian at rank 13 (of 15) at every waypoint, and R free along the path (least ‖X − R(X)‖ > 0). All 23 landed (Table 2, Appendix B). The paths run from 562 to 1558 rebuilt waypoints, with self-clearance between 0.537 and 1.564 bond lengths, least ‖X − R(X)‖ between 2.244 and 5.200, maximum residual 9.9e-14, and the Jacobian at rank 13 at every waypoint of every path.

This is Law C in matter. The improper mirror of space, which no rigid motion of the held chain can supply, is supplied by an internal loop one freedom up. (The five pairs at freedom 1 and the twenty-three at freedom 2 are different windows; the ladder is a statement about freedom, not about one tracked window.) A conformation reaches its mirror image after one lap of that loop and comes home after two.

What a join is. Along each path every gauche torsion changes hand and the path never touches the mirror-fixed set; after one lap the distance matrix has returned and the chain is its own mirror image, and after a second lap, along the reflected path, it is home. The quotient loop is the twisted clock of Section 2 in the shape of a molecule.

What it costs is a separate question. These are statements about kinematics — exact bond lengths and angles, zero-thickness bonds. A real chain pays for each passage through an eclipsed or contact barrier; the join shows that a way exists, and that energetic layer is OPEN. The notebook stores each path as every-twentieth waypoint and rebuilds it by projection at step 0.02, re-verifying ends, residual, rank and clearance (check 4.4b).

7. What it is

A study of readers. The record began with relation vocabularies read through symmetry readers and ended with chains read through distances and torsions. In between, every reader was a quotient and every advance was one of three moves — name the kernel, build the route back, prove the obstruction — and the discrete part of the kernel kept being one bit. Even the first witness carries it: its rational irreducibles come in ± pairs

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split by a central involution, and the labelled arena is a connected double cover on which that involution exchanges the two tokens (§ 81). The record's first decomposition was already by the two characters of a swap.

A ℤ/2 gauge structure. A held frame is a gauge choice, the sheet is the fibre, the odd loop is a holonomy of −1 — the defining data of a principal ℤ/2 bundle, the simplest gauge structure there is. Physics has this type in at least four places, and the record has met each: the spinor sign after one full turn (check 3.9); the exchange sign of identical particles in three dimensions (check 3.5; Leinaas–Myrheim); the sign change of a real electronic wavefunction carried round a conical intersection (Longuet-Higgins et al. 1958; Herzberg–Longuet-Higgins 1963), the ℤ/2 case of Berry's geometric phase, of the same type as the honeycomb holonomy of § 145; and conformational chirality undone by internal motion (Section 6). This is an identification of *type* and it stands; whether a given sector is physically any one of these is open.

§ 141, bound by type. § 141 left one item open: which physical interaction supplies the single net inversion. Reversed: name the gap (one swap, applied once, no fixed point); subtract the proper part (where the swap is a proper motion and the frame free, a rigid half-turn supplies it — the planar mirror is a half-turn of space); the remainder is the improper part, handedness in space, which no rigid motion supplies; it copies the same free involution one level up. **Bound per level, with one selector: is the swap proper among the motions available?** If proper with a free frame, the provider is the frame; if improper or the frame is held, the provider is an internal loop — in every case measured, available one freedom above the level at which the sheets could not be reached from each other (Section 6). Kinematically the inversion is supplied by a *path*, not an interaction; what an interaction supplies is the price of the path, and that stays open.

What it is not. It is not a derivation of physics from bits. Every identification above is of type; the values come from the material — bond angles, energies, rates, constants. The first matter number shows where type ends and values begin.

8. The first matter number: the averaged running product

Flory's generator matrices for the end-to-end distance multiply one bond transformation per torsion state along the chain — the § 160 running product $G_n = G_0 g_1 \cdots g_n$, weighted by the rotational-isomeric-state Gibbs measure, with C_∞ its second moment per bond. For independent torsions the average has the closed form $2(2+\sigma)/(3\sigma)$ (check 6.4); equal weights give 2, which is geometry alone (the tetrahedral cosine 1/3).

model (ideal tetrahedral geometry)	σ	ω	C_∞
equal weights	1	1	2.000000
413 K, gauche only	0.542507	1	3.124392
413 K, gauche and pentane	0.542507	0.086621	6.229676
413 K, pentane forbidden	0.542507	0	6.715345
300 K, gauche and pentane	0.430888	0.034471	7.200650

Table 3. The characteristic ratio of the ideal chain, with E_g = 2100 and E_p = 8400 J/mol (Nairn) and no fit (check 6.1). The weights are shown to six decimals; C_∞ is computed from the unrounded weights $\exp(-E/RT)$, and its sixth decimal moves with that rounding (§ 168). For comparison, Han, Jaffe and Yoon (1997) report $C_\infty \approx 7.9$ at 413 K from simulation, corroborated by small-angle neutron scattering on nearly monodisperse melts (7.8 ± 0.4), against the generally accepted 6.8 from intrinsic viscosity; the comparison is retrospective.

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The weights cannot close the gap. At 413 K, C_∞ falls monotonically as the pentane weight rises (check 6.3), and its ceiling with the pentane conformation forbidden is 6.7153 — below the neutron band. The residual lies outside the weights, in the chain geometry: opening the bond angle from 109.47° to the 112° used in RIS treatments of polyethylene (e.g. Briels) raises C_∞ to 6.886 at the same weights, by two independent methods (§ 168, checks 168.12–168.12b; a post-hoc route back, not a landing). One synthesis reviewed for this paper attributes the closing values (112° bond angle, 113° gauche minima) to Han–Jaffe–Yoon; their abstract does not say this, so the attribution is UNLINKED.

One observable cannot fix two energies. The trans fraction is 0.596664 at 413 K, and the Jacobian of (C_∞, trans fraction) in (E_g, E_p) is non-singular (det −0.004598 per (kJ/mol)²), so the two observables pin both energies locally (check 6.5). A matched measurement of the trans fraction is UNLINKED; the prospective test is to seal both readouts, with their temperature and geometry, before the data are opened.

9. Open, with the missing link named

- **B without fixed points at every width** (§ 143; runs to n = 22). Missing: an all-width argument.
- **Pivot-graph connectivity at every width** (§§ 145, 155). Missing: a proof, or Madras–Sokal matched to the honeycomb.
- **The ladder, as a theorem.** Measured: 5 of 5 not reached at freedom 1 (every traced branch closed), 23 of 23 joined at freedom 2. Missing: an exact real-algebraic proof that the freedom-1 components are disjoint, and a proof that mirror sheets always join one freedom above where they separate, or a counterexample.
- **The native address order.** τ and P bind on the sheet quotient as rotation and reflection, and the group they generate is now understood (Section 2). The native successor is not an element of that structure. Missing: how the ring's address order relates to ⟨τ, P⟩.
- **The energetic layer of every join**: cis passages, barriers, thickness, rates. Missing: an energy function sealed before use.
- **The physical phase of H_- and of the frame holonomy**, now typed as a $\mathbb{Z}/2$ holonomy but not identified.
- **Polyethylene**: a second observable sealed before its data; the geometry that closes 6.23 → 7.8; a proof that C_∞ is monotone in ω.
- **Interval-certified clearance along the joins.** The waypoints are verified and the step between them is below 0.03 bond lengths; a Lipschitz bound would certify the whole sweep.

10. Conclusion

Pulled all the way back, the NEXUS record is one fact read many times. A distinction that must continue is a swap. A reader that cannot see the swap loses one bit, and the bit comes home in one of two ways — it is held, or it takes a second lap. Moving up a level turns the swap into a half-turn whose own lift carries it again as a sign. Read in frequency, the swap's second character is the Nyquist frequency of its two-element group, and two laps home is the Nyquist count.

The record found this in a bit and a register, a sphere of readings, two tokens, a Hamiltonian, a planar chain and a tetrahedral one, and the size of a polymer coil. The sealed run of § 165 added the clearest instance yet: every mirror pair of tetrahedral conformations the tracer cannot connect at one freedom is joined at the next, through a loop that reaches the mirror image after one lap and comes home after two.

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The routes were fixed the whole time. Nothing changed them; everything that happened was a traversal of them, read at the moment of passage. That is what the record has been mapping — not a program, but the fixed graph a program would have to run on, and the one swap that is its shape.

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Appendix A — inventory of new results

Headline results by group; status is the ledger's. STANDS marks mathematics or a measured fact supplied from outside and recovered here; RUNS marks a computation that landed; DEAD END and OPEN as named.

result	where	status
a reader's number is ⟨m(V), f_H⟩; a zero has three causes	§§ 1–47	RUNS
vocabulary and reader are two non-substituting slots; interface width 6	§§ 24–26	RUNS
the minus sector is the Nyquist band of the central involution	§ 37	RUNS
the two-token exchange class is $\mathbb{Z}/2$, the non-planar braid torsion	§§ 73–75	STANDS; RUNS
exactly one homomorphism of the walk onto the two-state swap; turn = continue then NOT	§ 81	RUNS
+1 is NOT called recursively through the carry; swap twice is 2[z]=0	§ 84	STANDS; RUNS
the half-adder loses exactly w/2 bits; every fibre is one swap orbit	§ 99	RUNS
a name is a collision of two routes; a digest is a name, not unique data	§§ 100, 107	RUNS; DEAD END (routed)
capacity $2n^2$ with nothing supplied; $2\ell+1$ the surface, the 2 the bit	§ 118	RUNS; STANDS
the register stack sums to 2^{64}; layer n/2 is the half-turn fixed set of layer n	§ 117	RUNS
eight BBP lanes: seven constants and one zero route	§ 131	RUNS (exact)
the native partner has no fixed point; readings form a sphere, C the antipode	§§ 135–137	RUNS; STANDS

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result	where	status
the feedback filtration leaves the complement; $\tau^n = C$; the orbit law	§§ 139–141	RUNS; STANDS
which interaction supplies the single net inversion	§ 141	OPEN; bound by type in § 165
C is the one energy-preserving operator; $H = H_+ \oplus H_-$; holonomy −1	§§ 145–146	RUNS; proved
the planar mirror is a proper half-turn of space; all 46 locks open in space	§§ 152–155	STANDS; RUNS (≥ 0.977)
head-on to overhead is the running product with a held frame	§ 160	STANDS; RUNS
every planar contact is invisible at first order	§ 161	RUNS (226 contacts)
the 12-cycle crankshaft is locally rigid (exact PD certificate)	§ 162	proved
diamond exchanges with no search: 0, 12, 21, 213, 981	§ 163	RUNS
16 of 38 seven-bond exchanges connect; Schatzki's crankshaft an exact 120° turn	§ 163	RUNS; STANDS
polyethylene C∞ from the shape channel: 2, 3.12, 6.23 (413 K), 7.20 (300 K)	§ 164	RUNS; retrospective
the twisted ring is the binary cyclic group	§ 165	STANDS; RUNS (sealed)
the mirror-pair ladder: not reached at freedom 1 (tracer), joined at freedom 2, R free	§ 165	RUNS (sealed, 5/5 and 23/23); disjointness at freedom 1 OPEN
the counter as a tower of double covers; the register stack as a tower of fixed sets	§ 165	STANDS; RUNS
the twisted register and the native partner: rotation and reflection of one dihedral quotient; P is J corrected by C_1, not in $\langle \tau, J \rangle$ for $n \geq 3$	§ 166	STANDS; RUNS
two laps is the Nyquist count of the swap group; the 2n-step spectrum splits by parity, state on odd and reader on even harmonics	§ 166	STANDS; RUNS
P is affine over $\mathbb{F}_2$ exactly when the palindromes form a hyperplane ($n \leq 3$)	§ 166	RUNS (post hoc; the sealed prediction missed)
the group τ and P generate: $\lvert G_n \rvert = 2n \cdot 2^n$ ($n = 3$, $n \geq 5$; 64 at $n = 4$); at odd n the signed n-gon $\mathbb{Z}/2 \wr D_n$, faithful on the principal orbit; at even n an index-2 image and one central non-split swap	§ 167	STANDS (proof); RUNS ($n = 3..10$)
a chord $(P\tau^s)^2$ flips two sheet pairs on each orbit that meets a palindrome, or none; the orbits it acts on, in closed form; at odd width no chord cancels a harmonic	§ 168	STANDS (proof); RUNS (to $n = 16$; counts to $n = 20$)

Appendix B — the seals and the ladder

Seals. § 165 seal `73b2c9efd364e5b912c7513391935d24b8ff43199a3d349fba3793ad89388c78` (the ladder and the recursion laws), hashed 2026-10-24T12:46:46Z, before the run. Notebook seal C `1d5680ded94034480df16cf4b45518528bb85f288b4ac718e1291dc89fd2d68f` (the 27 notebook

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statements), hashed 2026-10-24T12:57:36Z, before the notebook ran. The lost originals were `65eec639…` (08:24:05Z) and `bfc5ed90…` (08:45:46Z). Version 2: seal 166 `155fd438e30a887b8d999a9e1bcd2865f2bdf83c4900f9389f38d311c2bce904` (the 39 statements of the corrected notebook), hashed 2026-10-24T23:07:43Z, before any of its code was written; it supersedes seal C's statement list. Version 3: seal 167 `0f904040093b3f9b49b173c5f81716646c493ef35524a286e3775beb0f9deb7a` (the ten statements of the group audit), hashed 2026-10-24T23:52:30Z, and seal 168 `caac6cd5d439b462a50af93bd6c63a434a75db40fee5b94ea6296c64e74ef466` (twelve statements, among them the bond-angle reproduction), hashed 2026-10-25T00:50:11Z, each before its code was written.

Table 2. The twenty-three native eight-bond mirror pairs, all joined (seal `73b2c9ef…`). Each path was rebuilt from its stored every-twentieth waypoints and re-verified; clearance is the least distance between non-adjacent bonds (zero thickness).

orbit (8 bonds)	A ↔ B	mirror	rank	clearance	‖X−R(X)‖
0	tttg−g+g− ↔ tttg+g−g+	xy	13	0.867	3.665
17	tttg−g+g+ ↔ tttg+g−g−	xy	13	1.433	3.796
46	g−g−g+ttg+ ↔ g+g+g−ttg−	yz	13	1.171	3.218
49	g−g−g+ttt ↔ g+g+g−ttt	yz	13	1.435	3.603
52	ttg−g−g+g+ ↔ ttg+g+g−g−	xz	13	1.193	3.510
56	g−tg+g+g−g+ ↔ g+tg−g−g+g−	xz	13	1.000	2.842
57	ttg−g−g+g− ↔ ttg+g+g−g+	xz	13	1.000	5.200
59	g−tg+g+g−g− ↔ g+tg−g−g+g+	xz	13	1.257	4.280
76	tg+ttg−g+ ↔ tg−ttg+g−	yz	13	0.904	2.654
80	tg+ttg−t ↔ tg−ttg+t	yz	13	1.564	4.452
114	g−g−g−g+g+g− ↔ g+g+g+g−g−g+	xy	13	1.000	4.602
123	g−g−g−g+g+g+ ↔ g+g+g+g−g−g−	xy	13	1.331	4.781
127	g+ttg+g−g− ↔ g−ttg−g+g+	xy	13	1.285	3.336
135	g−g+g−ttt ↔ g+g−g+ttt	xy	13	0.887	3.665
141	g−g+g−ttg− ↔ g+g−g+ttg+	xy	13	0.608	3.561
144	g−g+ttg−t ↔ g+g−ttg+t	xz	13	0.888	2.498
146	g−g+g−g−tg+ ↔ g+g−g+g+tg−	xy	13	0.809	2.244
150	g−g−g+g+tt ↔ g+g+g−g−tt	xy	13	1.252	3.619
155	g−g+g−g−tt ↔ g+g−g+g+tt	xy	13	0.537	4.387
157	g−g−g+g+tg− ↔ g+g+g−g−tg+	xy	13	1.257	4.286
168	g−g+ttg−g+ ↔ g+g−ttg+g−	xz	13	1.257	4.221
171	g−ttg−g+g− ↔ g+ttg+g−g+	xz	13	0.608	3.735
174	g−g+g+g−g−g− ↔ g+g−g−g+g+g+	xy	13	1.000	4.802

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Appendix C — corrections and provenance

- **Lost first build.** The in-progress § 164/§ 165 build — the first draft of this paper, the notebook, and an 86-minute run of the eight-bond joins — was lost when the computing environment was reset on 24 October. The diamond enumeration is deterministic, so the orbit counts and the mirror-pair indices reproduced exactly on rebuild; the predictions were re-sealed fresh (Appendix B) and re-run. The earlier digests stand as the dated record that the predictions preceded the run.
- **Scope.** The original work split the diamond result into "R3" (the five seven-bond pairs extended by one bond) and "P1" (the native eight-bond pairs). The extension is geometrically delicate — an extended window can land on an isolated lattice point — and is not reproduced here. The merged statement, the ladder, is cleaner and rests only on the trusted § 163 enumeration and tracer plus the manifold search: five pairs not reached at freedom 1, twenty-three joined at freedom 2.
- **Six hollow checks.** In the § 165 notebook six of the 27 checks did not test their claims. 3.1 compared the Euler-characteristic formula with itself; 5.1 tested that integer lattice points have integer squared distances, which says nothing about derivatives; 6.4 substituted $\sigma = 1$ into the formula it meant to derive. GPT found these three, and understated them. 1.7 never computed the character action it named; 2.3 used an orthogonal map that could be a reflection and checked neither torsions nor the rebuild; 4.1 checked properties of the matrix diag(1, −1, −1) rather than the mirror. A re-audit found the other three. They were weakened when the notebook was rebuilt after the reset; the lost originals were real. All six are replaced in the § 166 notebook by real computations — the homology of the complex of readings over $\mathbb{Q}$ and $\mathbb{F}_2$, a derivative-scaling test with a non-planar control, a symbolic derivation tied to the generator matrices, the character action and its SU(2) lift, a proper rotation with torsions and a rebuild, and the mirror of all 8188 turn words — and all six pass.
- **One bit (GPT).** Version 1 said a reader's kernel is always one bit. That is exact for the record's binary readers; continuous readers also have a continuous kernel, the frame. Corrected throughout: the discrete part is the one bit.
- **Separated (GPT).** Version 1 called the freedom-1 pairs separated. The tracer's non-reachability is computational, not a proof of disjointness; reworded.
- **Gemini's bridge.** The presentation $J\tau J = \tau^{-1}$, the order 4n and the rotation and reflection downstairs are right (checks 7.2, 7.4). Three parts are not: {I, P, C, PC} is not a subgroup of ⟨τ, J⟩ (P is not in it for $n \geq 3$; the Klein group there is {I, J, C, JC}); lap two of the principal orbit lies in the right half of the ring only for $n \geq 4$; and sampling the difference reader over the whole two-lap window does not restore the sheet (checks 7.3, 7.6, 7.11). The closing claim that τ and the native ring "are no longer two separate models" goes too far: τ and P bind on the sheet quotient, but the ring's address order is not conjugate to τ (check 7.7).
- **Sealed miss (§ 166).** Seal 166 predicted that P is not affine over $\mathbb{F}_2$ for n = 3..6. At n = 3 it is. There the palindromes form a hyperplane, and P — J composed with the complement on the palindromes — is affine exactly then. Logged as a miss (check 7.5); the corrected statement is post hoc (check 7.5b).
- **Rendering.** Version 1 wrote its formulas as Office Math objects, which some viewers show as blanks; version 2 writes them as plain text.
- **R2(g).** As first sealed, the SU(2) line turned the bond by half the intended angle per step; rerun with the intended step, the lift is −I after one full turn and +I after two (check 3.9).
- **SHA.** SHA is kept structural: a deterministic transformation in the binary domain whose digest is a many-to-one reader (an equivalence class of states, not an input). It can be run backward only given its full state; recovering an input from a digest alone is not attempted. Nothing here bears on any hash function.

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- **UNLINKED.** One synthesis reviewed for this paper attributes to Han–Jaffe–Yoon the claim that the measured C∞ requires a 112° bond angle and 113° gauche minima; their abstract does not say so, and their full paper was not read. A matched trans-fraction measurement for the two-observable test is likewise UNLINKED.
- **Version 3.** Version 2 listed the group τ and P generate as measured but not understood; § 167 closed it, and Section 2 now states the result. The 112° value in Section 8 was correct but had no reproduction in the companion notebook, and its source had been lost in the rebuild; it is now reproduced by two independent methods (§ 168) and sourced (Briels). Table 3's caption now says that its weights are rounded to six decimals while C∞ is computed from the unrounded weights; a sealed check in § 168 missed at the sixth decimal for exactly that reason, and is logged. The Han–Jaffe–Yoon comparison now reports what their abstract says, rather than placing all three values at 413 K. Passages written for the working exchange — an instruction quoted by date, "attached" material, corrections marked as the writer's own — are reworded for outside readers, and Table 2's columns are resized so that each mirror pair fits on one line.
- **How the work was done.** The computations, the notebooks and the text were produced with Claude (Anthropic) under the author's direction; GPT (OpenAI) and Gemini (Google) reviewed drafts and contributed results, which are credited where they are used.
- **Provenance.** Every number in the argument comes from the § 166 notebook's output, the sealed ladder run's output (out165.txt), the § 167 and § 168 notebooks, or the ledger sections cited. "Centre" keeps the series' British spelling.

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